References, Further Reading and Useful Tools

Free and publicly available sources for this course

1. Free textbooks

Source	Sections matching this course
OpenStax, *Astronomy 2e* (Fraknoi, Morrison, Wolff et al., CC BY 4.0) https://openstax.org/details/books/astronomy-2e	§2.1 The Sky Above, §2.2 Ancient Astronomy (Units 1–2) · §4.1 Earth and Sky, §4.2 The Seasons, §4.3 Keeping Time, §4.4 The Calendar, §4.5 Phases and Motions of the Moon, §4.7 Eclipses of the Sun and Moon (Unit 1) · §6.1 Telescopes, §6.2 Telescopes Today, §17.1 The Brightness of Stars, §17.2 Colors of Stars, §19.1 Fundamental Units of Distance, §19.2 Surveying the Stars (Unit 3)
NCERT *Physics* Class XII, Part II https://ncert.nic.in/textbook.php	Ch. 9 "Ray Optics and Optical Instruments" — telescopes and magnifying power (Unit 3)
Report of the Calendar Reform Committee (Govt. of India, 1955; chair M. N. Saha)	freely readable on the Internet Archive (archive.org) — background for Lecture 1.4
AAVSO Visual Variable Star Observing Manual https://www.aavso.org	estimating magnitudes by eye (Lecture 3.1)

2. Widely available books

- H. A. Rey, *The Stars: A New Way to See Them* (Houghton Mifflin) — the classic beginner's guide to the constellations.
- G. Consolmagno & D. M. Davis, *Turn Left at Orion* (Cambridge University Press) — what you really see in a small telescope.
- T. Dickinson, *NightWatch: A Practical Guide to Viewing the Universe*; T. Dickinson & A. Dyer, *The Backyard Astronomer's Guide* (Firefly Books).
- R. W. Sinnott, *Sky & Telescope's Pocket Sky Atlas*.
- S. J. O'Meara, *Deep-Sky Companions: The Messier Objects* (Cambridge University Press).
- S. Balachandra Rao, *Indian Astronomy: An Introduction* (Universities Press) — the Panchang and Indian astronomy.
- E. G. Richards, *Mapping Time: The Calendar and its History* (Oxford University Press).

3. Syllabi and olympiad resources

- Junior IOAA syllabus — https://ioaastrophysics.org/junior-ioaa/junior-ioaa-syllabus; IOAA syllabus — https://ioaastrophysics.org
- Indian National Astronomy Olympiad (INAO) and NSEA, conducted by HBCSE (TIFR) with IAPT — https://olympiads.hbcse.tifr.res.in

Students interested in the olympiad route should master Lectures 2.1–2.2 (coordinates and time) and 3.1, 3.3 (magnitudes and telescope optics).

© Vaibhav Yenare

P.T.O.

4. Data sources used in the notes

- Star positions and magnitudes: Stellarium catalogues (Hipparcos / Gaia based), J2000.
- Sun declinations, Zero Shadow Days, Moon phases and planet positions for the course period: computed with Astropy.
- Eclipses visible from India: https://www.timeanddate.com/eclipse/in/india; NASA Eclipse Website https://eclipse.gsfc.nasa.gov.
- Islamic calendar dates (1448 AH began 16/17 June 2026; Ramadan 2027 expected ~8 February): published astronomical predictions (e.g. Gulf News, Jordan News, 2026) — always confirm against local moon-sighting announcements.
- Zero Shadow Day outreach: Astronomical Society of India, Public Outreach & Education Committee — https://astron-soc.in/outreach/

5. Free software and apps

Tool	Use
Stellarium (https://stellarium.org)	free planetarium for PC/Mac/phone; sky cultures incl. Indian nakshatras, Arabic al-Sufi, Babylonian
Sky Map / SkySafari / Star Walk	phone sky apps — use night (red) mode
Heavens-Above (https://www.heavens-above.com), NASA "Spot the Station"	ISS and satellite passes
Light Pollution Map (https://www.lightpollutionmap.info)	find dark sites
Clear Outside, Windy, IMD	cloud forecasts
SEDS Messier Database (http://www.messier.seds.org); NASA Hubble Messier Catalog	information and images for the Messier objects
International Meteor Organization (https://www.imo.net)	meteor shower calendar

6. Places to visit in India

Birla planetaria (Kolkata, Hyderabad, Chennai, Jaipur), Nehru planetaria (Mumbai, Delhi, Bengaluru), NCSM science centres, Kodaikanal Solar Observatory, Vainu Bappu Observatory (Kavalur), ARIES (Nainital), GMRT (Khodad, near Pune) on open days, Jantar Mantar (Jaipur, Delhi, Ujjain), and dark-sky destinations such as the Hanle Dark Sky Reserve (Ladakh), Spiti, the Rann of Kutch and rural Western Ghats.